A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 25

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** Temperature inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Temperature inversion** → a layer of warm air above cooler air that traps pollutants	1
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** On night A, temperature falls with altitude – warm effluent gases are less dense / more buoyant than the surrounding air so they rise **1** As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting **1** On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) **1** Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly **1** The cold ground cools the air in contact with it, producing a cold dense layer near the ground **1** Low wind velocity means the air layers of different temperatures do not mix **1** Valley allows cold dense air to collect / drain into the low ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
01.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Smoke smogs (eg London 1952) **1** Photochemical smogs (eg Los Angeles, Mexico City) **1** Build-up of NOx / carbon monoxide / PM10 in cities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Total = 42 + 26 + 31 + 9 + 22 = 130 kW **1** 31 / 130 x 100 = 23.8% (accept 24%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1 Cavity wall insulation:** the foam or fibre traps air in small pockets so convection in the cavity is prevented and the low thermal conductivity of still air reduces conduction through the wall **1 Low-emissivity double glazing:** the still air or inert gas between the panes conducts heat poorly, and the low-emissivity coating reflects infrared radiation back into the room instead of letting it radiate out **1 Heat exchanger:** outgoing stale warm air passes next to, but does not mix with, incoming cold fresh air, so most of the heat is transferred to the incoming air and the building can be ventilated without losing that heat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each extra layer saves less heat than the one before, because the layers already in place carry most of the temperature difference, so the saving per unit thickness falls **1** Beyond a certain thickness the cost and the embodied energy of making and fitting more insulation are never repaid by the energy it saves **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** A large power station runs at higher temperatures and pressures and uses large efficient turbines, so it converts a greater proportion of the fuel energy into electricity **1** The heat rejected by a large station is low-grade and the station is usually too far from any user, so it is lost to a river or cooling tower **1** A CHP unit is in the building that needs the heat, so the heat rejected by its generator is piped straight to the heating and hot water, and 70 - 80% of the fuel energy is used rather than about 35% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **combined heat and power** → using the waste heat from generation for heating	3

Do not write outside the box

Question	Marking guidance	Marks
02.5	**Any 5 of 6** 1 mark each for any 5 of the 6 points below, in any order, maximum 5. **1** Occupancy sensors and programmable thermostats so that lighting and heating run only where and when people are present **1** LED lighting, which converts a far greater proportion of the electricity into light and much less into waste heat than filament lamps **1** Large south-facing windows and a thermally massive structure to make use of passive solar gain, with shading to prevent overheating in summer **1** Heating water on demand at the point of use, so no energy is wasted keeping a tank hot and none is lost in the pipe run **1** Choosing materials with low embodied energy, and re-using the existing structure rather than demolishing and rebuilding **1 Changes in behaviour:** turning equipment off rather than leaving it on standby, closing doors, and setting the thermostat a degree lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-235; AQA spec 3.3.4.2.2-3.3.4.2.3.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored **1** Hydrogen has a high energy density so it can be transported / stored for long periods **1** Released by combustion for heat, industrial processes or steam turbines **OR** by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water **1** Can also be used as a vehicle fuel or converted to methane / ammonia **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
05.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks **1** Liquefied at very low temperature - energy for refrigeration and compression **1** Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy **1** Conversion to ammonia - some energy needed but less than the alternatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging **1** A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **fuel cell** → a cell that produces electricity from hydrogen and oxygen	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	1 mark for the point below. **1** Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **intermittent** → only generates when the wind blows or the Sun shines / supply is not continuous **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive **1** Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected **1** Damage to other cell parts can stop the cell functioning or kill it – low damage may be repaired by the intact nucleus **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Acute:** effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) **1 Chronic:** effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1 Exposure:** absorption of ionising radiation from a source the person is close enough to **1 Contamination:** physically carrying radioactive material on or in the body, so exposure continues wherever the person goes – sealed sources can expose but not contaminate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Sieverts = grays x radiation weighting factor (20 for alpha) **1** 0.002 x 20 = 0.04 Sv (40 mSv) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Sv) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.5	1 mark for the point below. **1** The activity of a source (number of decays per second) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **becquerel** → one radioactive decay per second	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** The scatter / variability of results around each mean – about 68% of values lie within one SD of the mean **1** The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** t-test **1** Comparing two means of measured (continuous) data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 30% probability that the difference occurred by random chance **1** Above the 0.05 (5%) level so the difference is not significant – the null hypothesis (no difference) is accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Two correlated variables may both be caused by a third variable that was not investigated **1** Significance only shows the result is unlikely to be chance – a mechanism / controlled experiment is needed for causation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together • **causal** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
07.5	**1 mark each** 1 mark for each of the 2 points below. **1** Increase the number of replicates (reduces the effect of variability) **1** Standardise other variables (soil, water, light) more closely to reduce scatter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
08.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
09.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 13 / 480 = 0.0271 years **1** 0.0271 x 365 **1** = 10 days (accept 9.9-10) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (days) is already printed on the answer line. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1 340 000 / 425 = 3153 years (accept 3100-3200) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (years) is already printed on the answer line. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • A bare wrong answer with no working.	1
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** It has a very short residence time / very rapid transfer rate (480×10^{15} dm^3 per year) **1** So it moves large volumes of water each year, delivering precipitation to land – small reservoirs with fast transfers respond quickly to change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** The ocean is so large that an added or removed volume is a tiny proportion of the total **1** The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) **1** But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface • **transpiration** → water lost as vapour from the leaves **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	
10.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 4.2 / 0.7 = 6 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low **1** In winter days are very short / the Sun does not rise, so light levels are extremely low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** Reflectors can only focus parallel rays of direct sunlight onto the absorber **1** Clouds, smoke and dust scatter the light so it cannot be focused and intensified **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	1 mark for the point below. **1** Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
10.5	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Problems:** intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • **Photothermal:** evacuated tubes reduce conduction / convection losses – thermal stores (water, sand, concrete, molten salt) allow later use – passive solar architecture, brise soleil, light tubes – heat pumps deliver up to four times the energy used • **Photovoltaic developments:** multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • **Integration:** grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • **Evaluation:** strong in sunny deserts / low latitudes, weaker in high latitudes in winter – cost falling – manufacturing impacts – no single technology solves all problems – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **pumped storage** → water pumped uphill when demand is low and released to generate at peak **Do not credit** • 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	
	Model answer - Level 3 (9 marks)	
	Solar power's problems are intrinsic: it is intermittent with daily and seasonal cycles, cloud makes it unpredictable and scatters light so it cannot be focused, its low energy density demands large collector areas, the angle of incidence keeps changing, cleaning in deserts uses scarce water, manufacturing PV produces toxic wastes such as silicon tetrachloride and cadmium, and PV efficiency falls as cells heat up.	
	New technology addresses most of these. Evacuated-tube photothermal panels cut conduction and convection losses, and thermal stores of water, sand, concrete or molten salt hold heat for later; passive solar design, brise soleil and light tubes reduce demand; heat pumps deliver up to four times the energy used to run them. Multijunction PV cells absorb more wavelengths, moth-eye anti-reflective surfaces stop the 30% of light a smooth cell reflects, PV-thermal hybrids cool the cell and use the heat, transparent PV turns windows into generators, heliostats keep panels at the optimum angle and nanohydrophobic surfaces self-clean. Concentrating solar power with molten salt stored at 550 °C runs steam turbines when electricity is wanted, directly overcoming intermittency, and grid-connected farms, rooftops that avoid land conflicts, batteries and pumped storage integrate the output.	
	Even so, no technology changes the fact that winter insolation at high latitudes is tiny, that CSP needs cloudless skies, or that manufacturing has impacts. Solar can dominate supply in sunny, low-latitude regions and contribute strongly elsewhere in summer, but overcoming its problems fully depends on storage and a diverse energy mix rather than on solar technology alone.	
	Weaker answer - Level 1 (2 marks)	
	Solar power has problems because it does not work at night or when it is cloudy, and solar panels are not very efficient. New technologies can help. Better solar panels are more efficient. Solar power can be stored in batteries for use at night. Mirrors can concentrate the sunlight to make it hotter. So new technologies can overcome most of the problems with solar power.	
	What makes the difference	
	• Level 1 identifies two problems and three vague technologies; Level 3 names each technology and explains which problem it solves and how.	
	• Level 3 acknowledges what technology cannot fix (high-latitude winters, cloud for CSP, manufacturing impacts) - the 'extent' in the question.	
	• Numbers (30% reflection, 550 °C, four-fold heat-pump output) show detailed knowledge applied to the argument.	

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
11.1	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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Question	Marking guidance	Marks
11.2	**Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Hydrological cycle processes and reservoirs – residence times – dynamic equilibrium – increasing demand (population, affluence, irrigation, industry) • **Human impacts on transfer rates:** deforestation, agriculture, urbanisation, climate change, abstraction – consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • **Case studies of unsustainable exploitation:** Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • **Sustainable strategies:** metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair – new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis – pollution control (sewage, AMD, buffer strips)	25
	• Links to thermohaline circulation and climate – energy costs – circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	

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